Stage 2 — Check-your-understanding Questions

Q1. A Caputo derivative by hand

For p > 0, the Caputo power rule is:

```
C D^alpha(t^p) = [Gamma(p+1)/Gamma(p+1-alpha)] t^(p-alpha).
```

Given Gamma(1/2) = sqrt(pi) and Gamma(z+1) = z Gamma(z).

For t, set p = 1 and alpha = 1/2:

```
Gamma(2) = 1; Gamma(3/2) = (1/2)Gamma(1/2) = sqrt(pi)/2.

C D^(1/2)(t) = [1/(sqrt(pi)/2)] t^(1/2) = 2 sqrt(t)/sqrt(pi).
```

The half-order derivative of t is not constant; its time exponent decreases by 1/2, not 1.

For t squared, set p = 2 and alpha = 1/2:

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Gamma(3) = 2; Gamma(5/2) = (3/2)Gamma(3/2) = 3 sqrt(pi)/4.

C D^(1/2)(t^2) = [2/(3 sqrt(pi)/4)] t^(3/2)

= 8 t^(3/2)/(3 sqrt(pi)).
```

At alpha = 1:

```
C D^1(t^p) = [Gamma(p+1)/Gamma(p)] t^(p-1) = p t^(p-1).

C D^1(t) = Gamma(2)/Gamma(1) = 1.

C D^1(t^2) = [Gamma(3)/Gamma(2)] t = 2t.
```

Q2. Why raise the rates to the power alpha?

Let t_week = t_day/7 and beta_week = 7 beta_day.

For the dimensionally consistent combination:

```
(beta_week)^alpha (t_week)^alpha

= (7 beta_day)^alpha (t_day/7)^alpha

= 7^alpha beta_day^alpha t_day^alpha / 7^alpha

= beta_day^alpha t_day^alpha (unchanged).
```

For the uncorrected combination:

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beta_week (t_week)^alpha = 7 beta_day (t_day/7)^alpha

= 7^(1-alpha) beta_day t_day^alpha (not unchanged).
```

At alpha = 1/2, the latter changes by sqrt(7), approximately 2.6458. Raising rates to alpha gives coefficients units of time^(-alpha), matching the Caputo derivative; predictions should not depend on whether time is expressed in days or weeks.

Q3. Same growth rate, different R0

Using r = 0.25 per week and sigma = gamma = 1 per week:

```
R0(alpha) = (1+(r/sigma)^alpha)(1+(r/gamma)^alpha)
= (1+0.25^alpha)^2.
```

At alpha = 1:

```
0.25^1 = 0.25; R0(1) = (1.25)^2 = 1.5625.
```

At alpha = 0.6:

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0.25^0.6 = exp(0.6 ln(0.25))
= exp(0.6 x -1.38629436)
= exp(-0.83177662) = 0.43527528.
R0(0.6) = (1+0.43527528)^2 = 2.06001513.
```

Percentage increase relative to the alpha = 1 estimate:

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[(2.06001513 - 1.5625)/1.5625] x 100 = 31.84097%.
```

Because r < sigma and r < gamma, the direction agrees with Proposition 10: R0 decreases as alpha increases. Identical observed growth rates can therefore yield different reported R0 values; a paper must identify the fractional model and alpha used.

Q4. The positivity proof

The projection P onto the feasible region makes f composed with P globally Lipschitz, ensuring a unique solution of the modified problem. The constant kappa = beta^alpha + sigma^alpha + gamma^alpha + mu^alpha is chosen so G(x) = (f composed with P)(x) + kappa x is non-negative for x >= 0. Since E_alpha(-x) > 0 and E_(alpha,alpha)(-x) >= 0, the variation-of-constants formula maps non-negative functions to non-negative functions. Every Picard iterate is therefore non-negative, and uniform convergence gives the non-negative limit x-tilde(t). The proof does not claim C D^alpha x >= 0 implies x is increasing, because this implication is false in general for a single Caputo order (Remark 1).

Q5. The AHM threshold

Proposition 8 gives:

```
T = 1/[gamma_c + tau^(-alpha) gamma_c^(1-alpha)].
```

For tau = 1 week, alpha = 0.5, gamma_c = 0.25 per week:

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tau^(-alpha) = 1^(-0.5) = 1.
gamma_c^(1-alpha) = 0.25^0.5 = sqrt(0.25) = 0.5.
T = 1/[0.25 + 1(0.5)] = 1/0.75 = 4/3 weeks
= 1.3333 weeks.
```

As gamma_c approaches zero from above:

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T = 1/[gamma_c + sqrt(gamma_c)] -> infinity.
```

Since R0 = omega S0 T, an infinite mean infectious period means any positive transmission omega (with S0 > 0) exceeds the growth threshold: no positive minimum transmission rate remains. Remark 3 contrasts the infinite integral of the AHM survival kernel when gamma_c = 0 and alpha < 1 with the finite integral 1/gamma^alpha of the Caputo infectious-compartment retention kernel; this is why the Caputo model retains a finite threshold.